

Multiple Choice (10 marks)

1. Life on Earth originated from
 - (a) We don't know for sure.
 - (b) aliens (panspermia).
 - (c) comets.
 - (d) the Earth's oceans.
2. What is true about the rotation of a solid disk?
 - (a) The velocity is the same on every point on the disk
 - (b) The part closer to the axis has a longer period
 - (c) The period of rotation is the same on every point on the disk
 - (d) The part closer to the axis has a shorter period
3. A comet of mass m impacts the earth (mass M radius R) at the minimum impact speed. What is the expression for the total energy released in the impact?
 - (a) $m \cdot v$
 - (b) $0.5 \cdot m / (R^3)$
 - (c) $0.5 \cdot m \cdot (2 GM/R)$
 - (d) $0.6 \cdot G(M^2)/R$
4. The distance between the Earth and the star Altair is one million times greater than the distance between the Earth and the Sun. How far is Altair from the Earth?
 - (a) 9.3×10^{13} meters
 - (b) 9.3×10^{10} meters
 - (c) 1.5×10^{14} meters
 - (d) 1.5×10^{17} meters
5. What do meteorites reveal about the solar system?
 - (a) They reveal that the early solar system consisted mostly of hydrogen and helium gas.
 - (b) They reveal that meteorites are much older than the comets and planets.
 - (c) They reveal that the age of the solar system is approximately 4.6 billion years.
 - (d) They reveal that the solar system once contained 10 planets.

True / False (10 marks)

Answer True or False

6. True or False: The Sun's corona is too diffuse, making it difficult to observe except during total solar eclipses when the photosphere is obscured.
7. True or False: The ability to breathe oxygen is not an essential requirement for life as we know it, as many organisms thrive in anaerobic environments.
8. True or False: Jupiter's magnetic field is approximately 20,000 times weaker than Earth's magnetic field.
9. True or False: A type-Ia supernova typically originates from massive stars in young galaxies undergoing rapid stellar formation.
10. True or False: Earth is the only planet in the solar system currently exhibiting active plate tectonics.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the methods through which scientists have determined that most meteorites originate from the asteroid belt, focusing on spectral analysis and its implications.
12. Discuss the significance of the findings from the Martian meteorite ALH 84001 and analyze why amino acids with a preferred chirality were not considered evidence for past life.

End of Paper